Few-Shot Radar Annotation

Goals:

- Develop high-accuracy method(s) to detect, locate, and classify small and large objects in radar sweeps in any scenario.

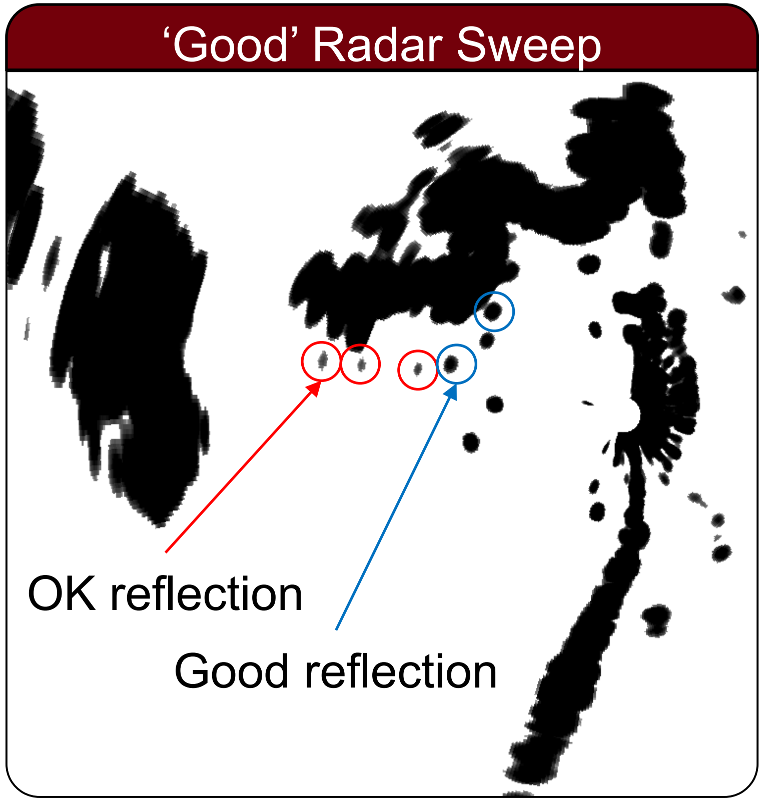
Progress Since Last QPR:

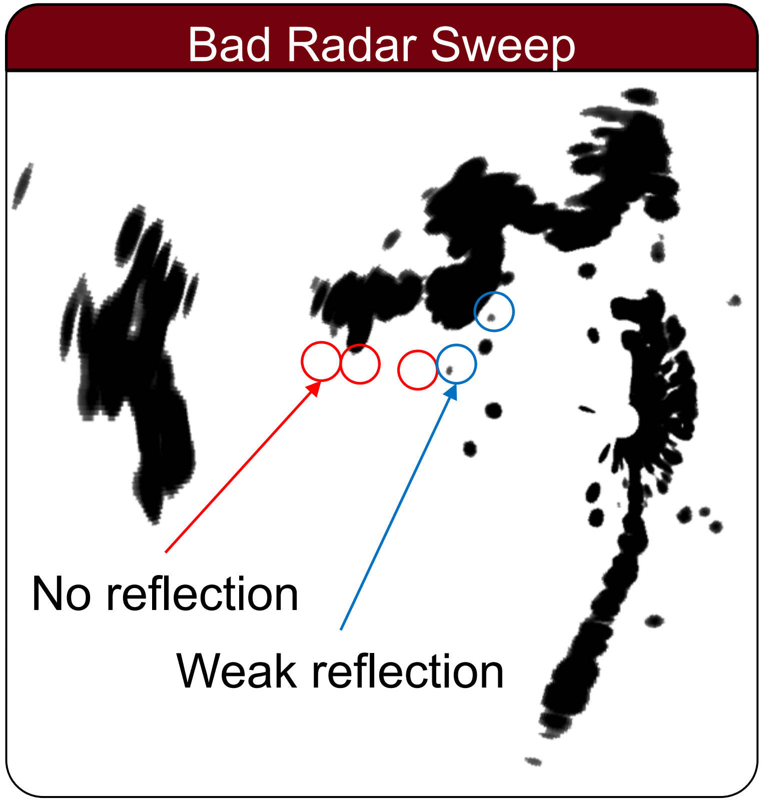
- Data capture trips across 9 field sites and 3 lakes, collecting over 100,000 sweeps
- Improved collection process; developed data storage method
- Explored HDR, radar characteristics, Standard tracking models, ML models, and Large Models (billions of params)
- Developed customizable non-electromagnetic simulations for radar sweeps

JC Vaught: MS Student (USC)
Lead: Dr. Doug Cahl (USC)

Challenges in radar annotation methods

- Small objects are invisible to consumer-grade radar in sea clutter
- Consumer-grade radar operates with limited 4-bit dynamic range
- Radar gain requires constant readjustment



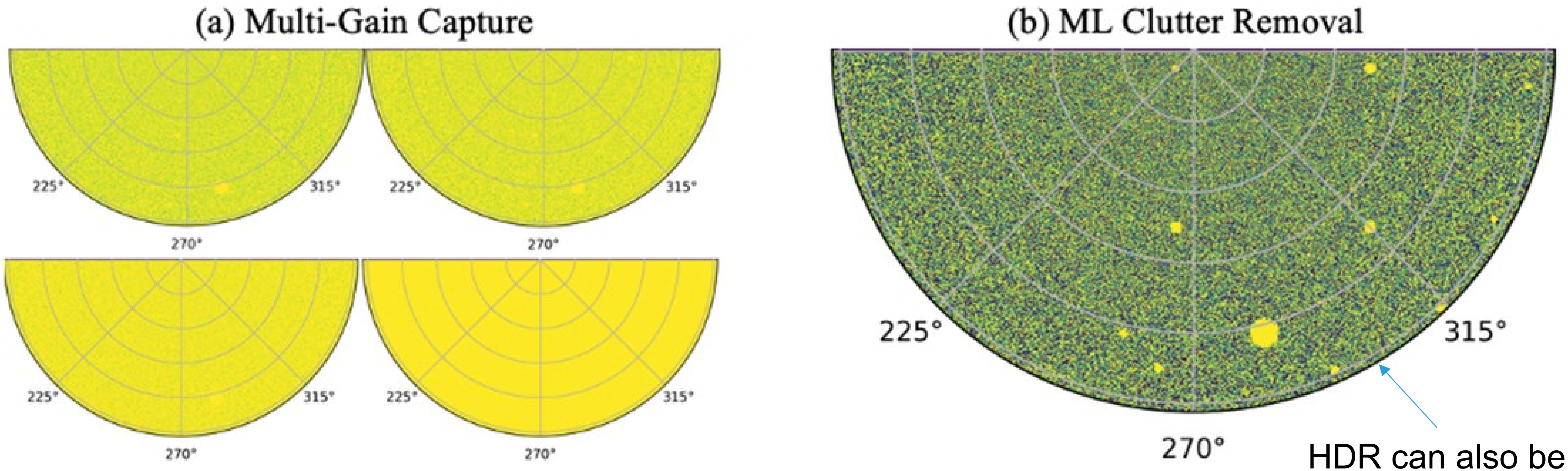


JC Vaught: MS Student (USC)
Lead: Dr. Doug Cahl (USC)

Overview of HDR & clutter removal

Tossing ‘cluttered’ frames into a sufficiently intelligent algorithm, a clear picture can be extracted.

Apply this idea across time(temporal space), and object ‘class’ can be derived



HDR can also be applied at this stage

JC Vaught: MS Student (USC)
Lead: Dr. Doug Cahl (USC)

Objectives and Results

Overview of Deliverables

- **Multi-Gain HDR Radar Software**
  - Combine multiple gain sweeps for extended dynamic range
- **Self-Supervised Learning**
  - Train detection models with **minimal** human annotation
- **Open Dataset**
  - Create an open dataset for maritime radar ML research

Engineering Targets

- **Classification & Detection:**
  - >95% P_d with 10^{-2} P_{fa}
- **Labelling Target:**
  - <5 labeled frames per site for few-shot learning
- **Dataset Target:**
  - 200+ unique collection sites with 2 hrs of data at each under a variety of weather conditions

JC Vaught: MS Student (USC)
Lead: Dr. Doug Cahl (USC)

CUI

Current Progress (May-August)

(MAY) Foundation

- Theoretical framework development
- Multi-gain concept introduction
- Initial hardware integration
- CFAR analysis and implementation

(JUNE-JULY) Data Collection

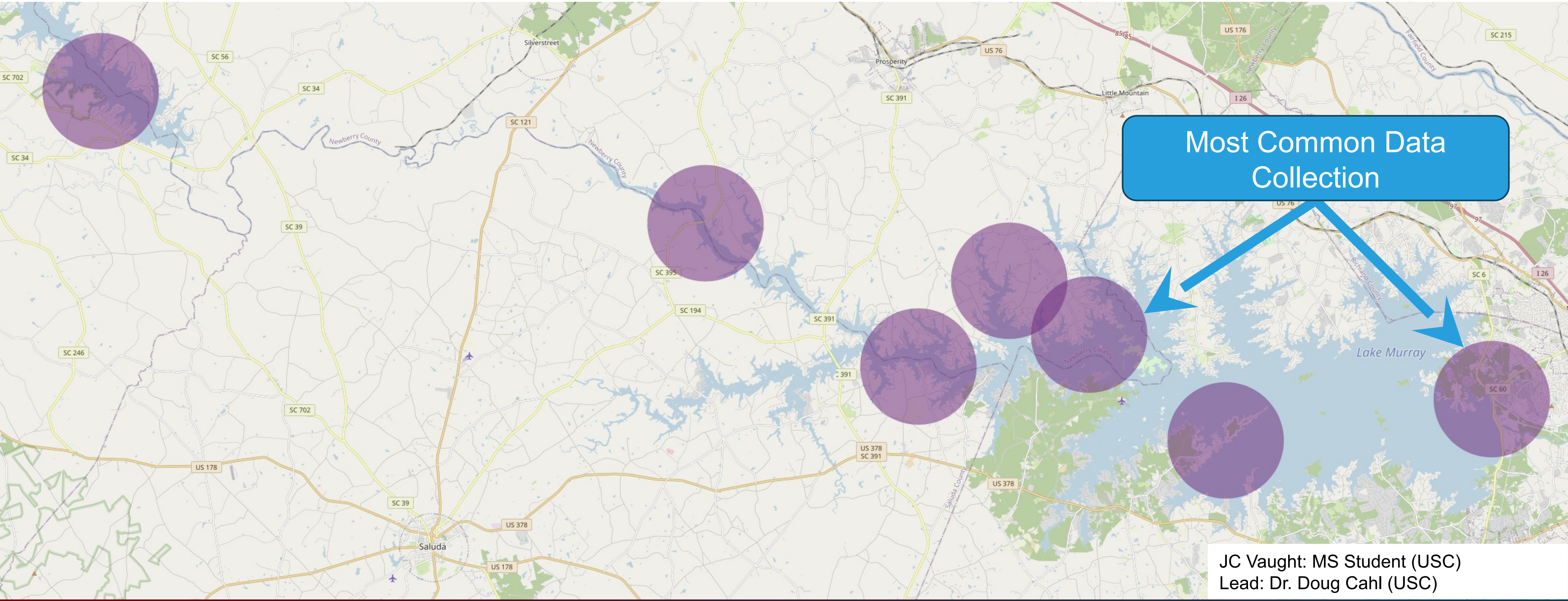
(AUGUST) ML Testing

- HDR system validation
- Tracking algorithms implemented
- Radar simulation development
- Corner reflector characterization

JC Vaught: MS Student (USC)
Lead: Dr. Doug Cahl (USC)

Data Collection Summary

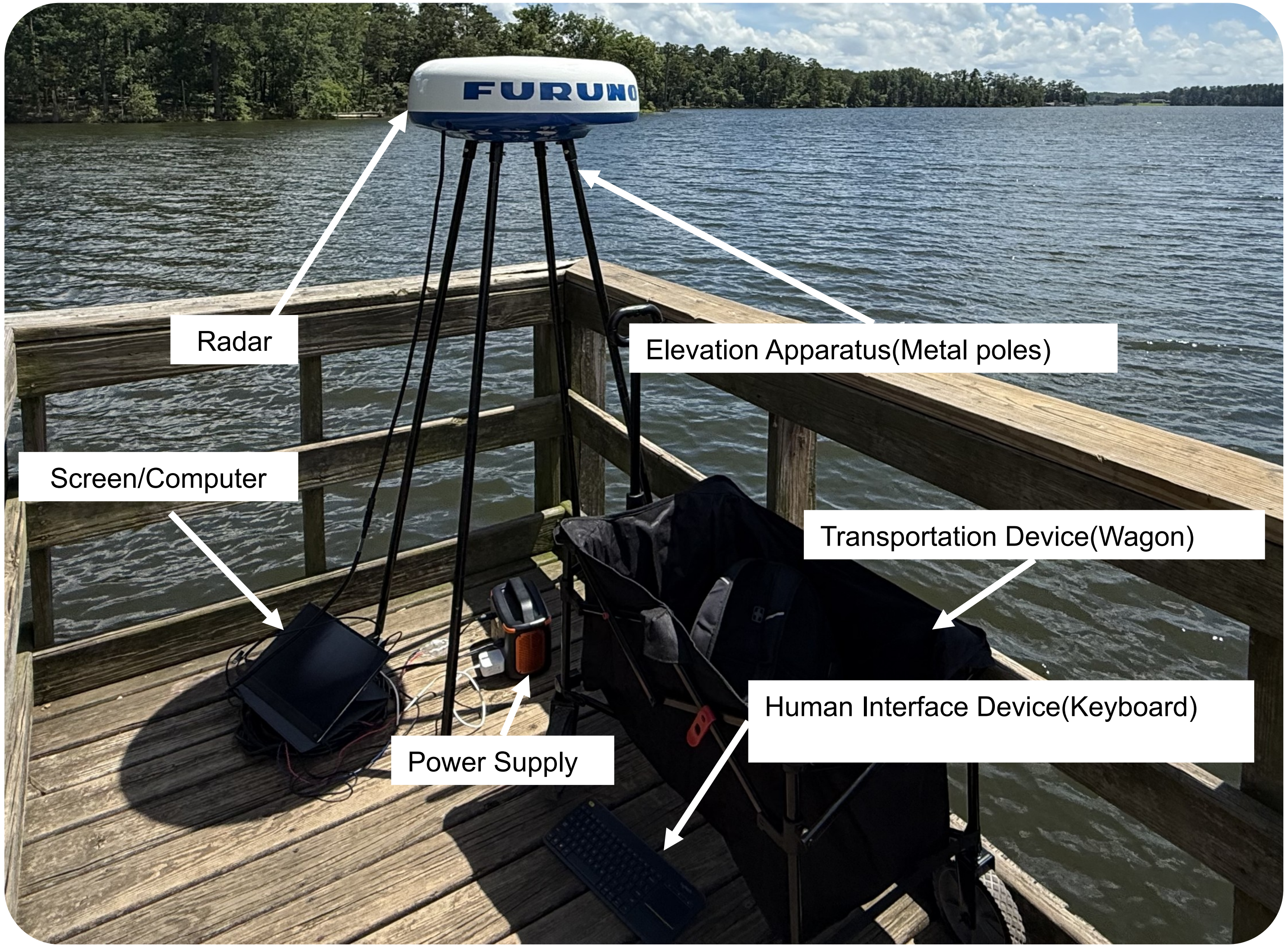
8 collection sites(2 not shown); 60,000 frames; 200+GB raw data



JC Vaught: MS Student (USC)
Lead: Dr. Doug Cahl (USC)

Collection Improvements

- 75% reduction in primary codebase (2000 to 500 lines)
- >90% reduction in deployment
- Boot time: 4-6 minutes to <20 seconds
  - (total setup <5 minutes)
- Enhanced reliability and portability



JC Vaught: MS Student (USC)
Lead: Dr. Doug Cahl (USC)

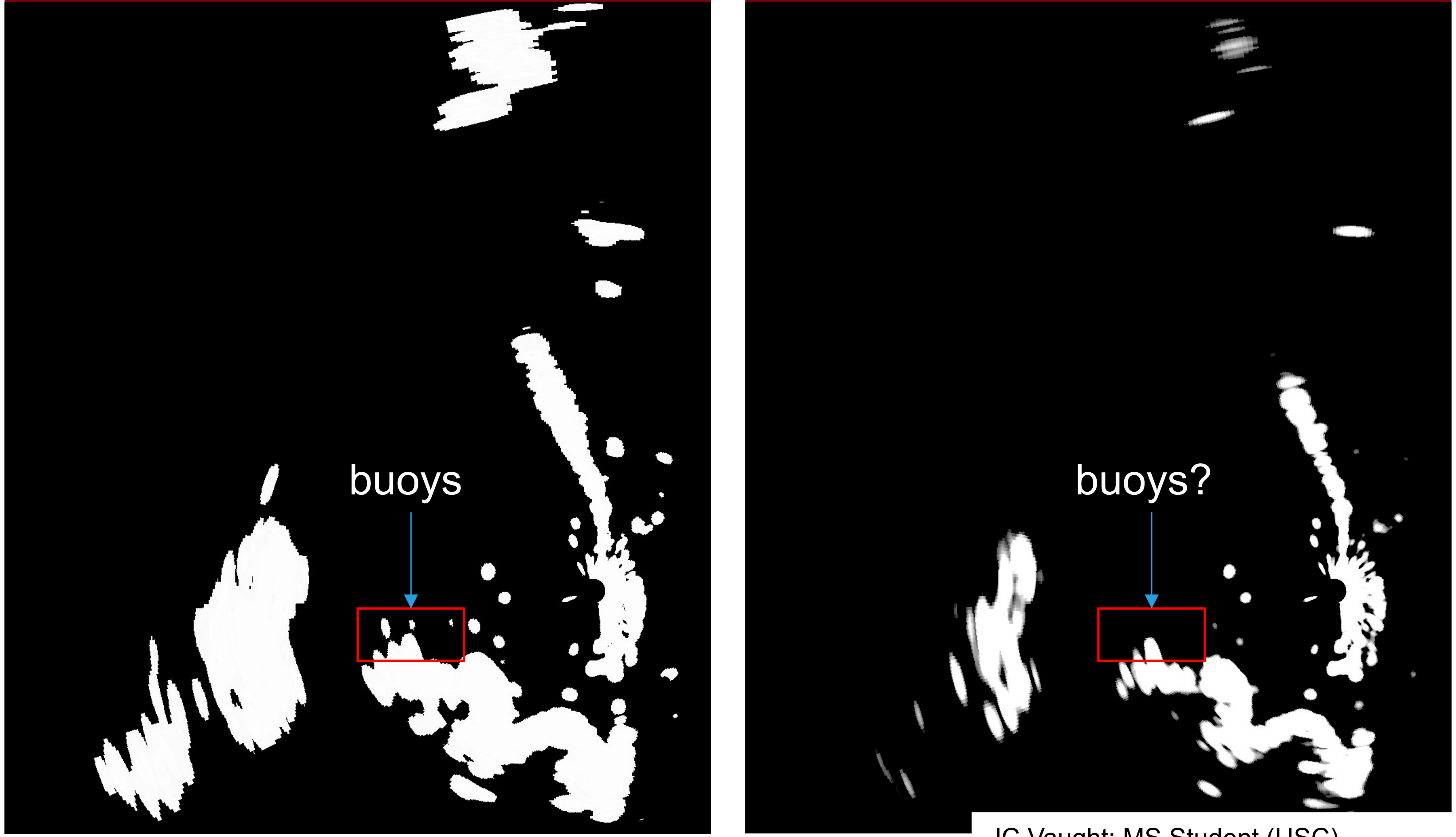
High Dynamic Range Implementation

Analysis & Comparison

- **Robertson Method** (chosen) Sigmoid response, preserved faint & bright echoes
- **Mertens Merge**: Good performance, secondary choice
- **Debevec**: Standard HDR, moderate results
- **Stacking Method**: In-house method, underperformed

HDR Sweep

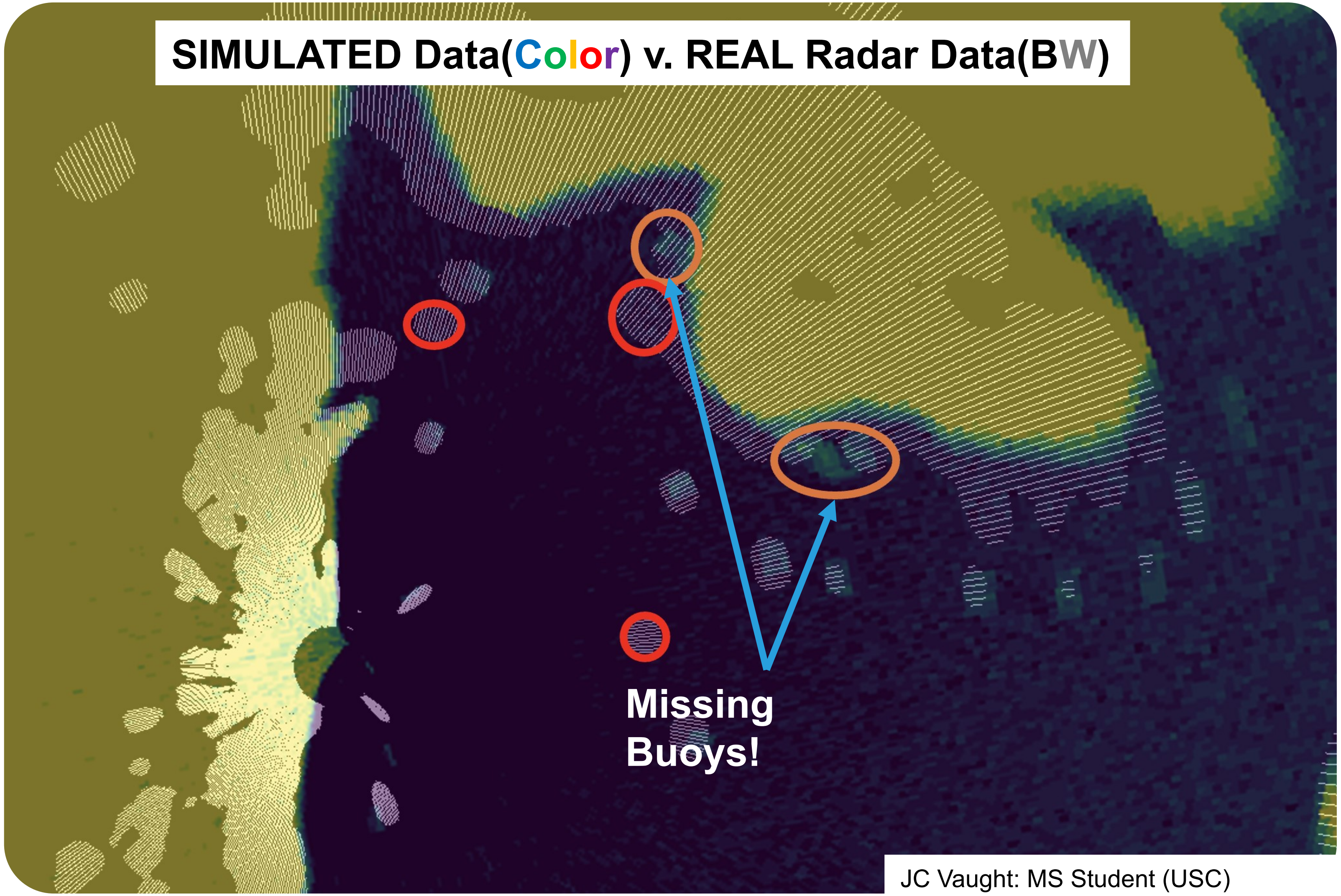
Single Sweep



JC Vaught: MS Student (USC)
Lead: Dr. Doug Cahl (USC)

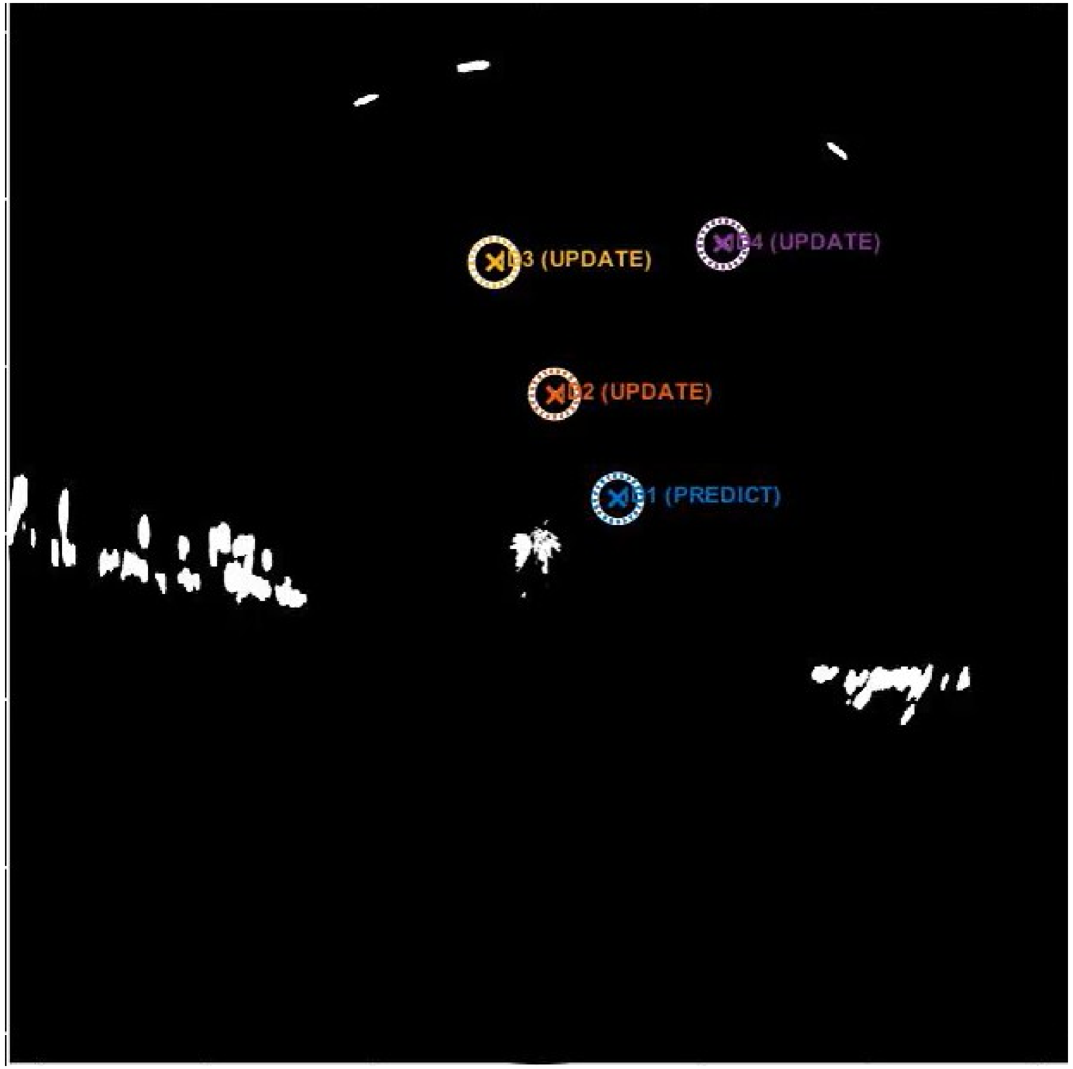
Geographically Accurate Testing Environment

- K, log-normal, Weibull, Rayleigh distributions for clutter
- Automatic shoreline data from public sources
- Cell-by-cell power map generation
- External antenna pattern tables & STC curves
- Quantization matching radar front-end (low-bit)



JC Vaught: MS Student (USC)
Lead: Dr. Doug Cahl (USC)

Manual tracking



Automatic tracking

JC Vaught: MS Student (USC)
Lead: Dr. Doug Cahl (USC)

Summary & Future Work

Completed Novel (=*Publishable*) Work

- Shannon entropy optimization replacing subjective gain adjustment
- HDR approach applicable to low-bit radar systems

Future Work/Currently In-Progress

- Geographically accurate radar simulation for controlled testing
- Large Model testing for masked auto-encoder(MAE) pipeline
- Multi-Pulse HDR system
- Temporal MAE for few-shot maritime object detection
- CNN + Kalman Object Tracking
- Few-Shot Radar Sweep Segmentation Models

JC Vaught: MS Student (USC)
Lead: Dr. Doug Cahl (USC)